Brain Tumor Methylation Reference Service

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DNA METHYLATION-BASED CNS TUMOR CLASSIFIER (v12.5) — FINAL REPORT

PATIENT	SPECIMEN	ORDERING
WALKER, AIDEN T. MRN MRN-7720193 8 y / Male	Reference accession BTMRS-26-M-1014 Client PED-26-N-00882 FFPE, B3 DNA extracted 2026-03-22	Children's Medical Center of the Lakes Tan, L., MD Received 2026-03-20 Reported 2026-04-02

CLASSIFIER RESULT

Methylation family: Medulloblastoma
Methylation class: MB, SHH-activated (subgroup 3)
Calibrated score: 0.94 (threshold for classification: ≥ 0.90)
TP53 status (by methylation-array model): Wild-type pattern

METHODOLOGY

Genomic DNA was extracted from FFPE tumor tissue (cassette B3). DNA (500 ng) was bisulfite-converted and assayed on the Illumina Infinium MethylationEPIC v2.0 array (~935,000 CpGs). Raw IDAT files were processed by an in-house pipeline (preprocessNoob, BMIQ, batch-effect correction) and the resulting beta-value matrix was submitted to the CNS Tumor Methylation Classifier (v12.5). A random-forest classifier assigns one of 184 methylation classes; calibrated scores derived via Platt scaling on an independent validation cohort.

TOP CLASSIFICATION CANDIDATES (calibrated scores)

Methylation class	Family	Score
MB, SHH-activated (subgroup 3)	Medulloblastoma	**0.94**
MB, SHH-activated (subgroup 1, infant)	Medulloblastoma	0.03
MB, SHH-activated (subgroup 4, TP53-mut)	Medulloblastoma	0.02
MB, Group 3	Medulloblastoma	0.01
MB, Group 4	Medulloblastoma	0.00
MB, WNT-activated	Medulloblastoma	0.00

The top class — MB, SHH-activated (subgroup 3) — receives a calibrated score well above the 0.90 threshold. The TP53-mutant SHH-activated subgroup ranks at 0.02, consistent with the molecular report's TP53 wild-type call.

COPY-NUMBER PROFILE (FROM ARRAY)

Array-derived copy-number profile is largely balanced. No MYC (8q24.21) or MYCN (2p24.3) amplification. No i(17q). No focal 9q22 (PTCH1 locus) deletion at array resolution (consistent with the somatic LOF point mutation reported on the NGS panel). The copy-number landscape is typical of MB, SHH-activated, in the school-age cohort.

INTERPRETATION

The methylation profile classifies this case as **MB, SHH-activated (subgroup 3)** with a calibrated score of 0.94 (high confidence). The methylation-based TP53 prediction model calls a wild-type pattern, concordant with the Sanger-confirmed TP53 wild-type result on the molecular report. Combined with the morphology (classic medulloblastoma), the IHC pattern (GAB1+, YAP1+, β-catenin cytoplasmic), and the molecular findings (PTCH1 LOF, SHH RNA signature), the case represents medulloblastoma, SHH-activated, TP53-wildtype, in the prognostically favorable molecular-risk category. WHO CNS5 grading: all medulloblastomas are CNS WHO grade 4 by definition.

LIMITATIONS

The methylation classifier is intended for adjunctive use alongside histopathology and molecular testing. Calibrated scores below 0.90 should not be used for classification. Tumor purity, array-batch effects, and age-related methylation drift can influence performance. The TP53 prediction is array-based; orthogonal confirmation by sequencing (reported separately) is recommended for clinical decision-making.

Electronically signed: Hofmann, Kerstin H., PhD, Director (Neuropathology Bioinformatics), and Adebayo, Tunde O., PhD, Senior Bioinformatics Scientist — 2026-04-02.